# Szpiro's conjecture

In number theory, **Szpiro's conjecture** relates to the conductor and the discriminant of an elliptic curve. In a slightly modified form, it is equivalent to the well-known *abc* conjecture. It is named for Lucien Szpiro, who formulated it in the 1980s. Szpiro's conjecture and its equivalent forms have been described as "the most important unsolved problem in Diophantine analysis" by Dorian Goldfeld,[1] in part to its large number of consequences in number theory including Roth's theorem, the Mordell conjecture, the Fermat–Catalan conjecture, and Brocard's problem.[2][3][4][5]

**Modified Szpiro conjecture**

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|---|---|
| **Field** | Number theory |
| **Conjectured by** | Lucien Szpiro |
| **Conjectured in** | 1981 |
| **Equivalent to** | abc conjecture |
| **Consequences** | Beal conjecture<br>Faltings's theorem<br>Fermat's Last Theorem<br>Fermat–Catalan conjecture<br>Roth's theorem<br>Tijdeman's theorem |

## Original statement

The conjecture states that: given $\varepsilon > 0$, there exists a constant $C(\varepsilon)$ such that for any elliptic curve $E$ defined over $\mathbf{Q}$ with minimal discriminant $\Delta$ and conductor $f$,

$$|\Delta| \leq C(\varepsilon) \cdot f^{6+\varepsilon}.$$

## Modified Szpiro conjecture

The **modified Szpiro conjecture** states that: given $\varepsilon > 0$, there exists a constant $C(\varepsilon)$ such that for any elliptic curve $E$ defined over $\mathbf{Q}$ with invariants $c_4$, $c_6$ and conductor $f$ (using notation from Tate's algorithm),

$$\max\{|c_4|^3, |c_6|^2\} \leq C(\varepsilon) \cdot f^{6+\varepsilon}.$$

### *abc* conjecture

The *abc* conjecture originated as the outcome of attempts by Joseph Oesterlé and David Masser to understand Szpiro's conjecture,[6] and was then shown to be equivalent to the modified Szpiro's conjecture.[7]

## Consequences

Szpiro's conjecture and its modified form are known to imply several important mathematical results and conjectures, including Roth's theorem,[8] Faltings's theorem,[9] Fermat–Catalan conjecture,[10] and a negative solution to the Erdős–Ulam problem.[11]

## Claimed proofs

In August 2012, Shinichi Mochizuki claimed a proof of Szpiro's conjecture by developing a new theory called inter-universal Teichmüller theory (IUTT).[12] However, the papers have not been accepted by the mathematical community as providing a proof of the conjecture,[13][14][15] with Peter Scholze and Jakob Stix concluding in March 2018 that the gap was "so severe that … small modifications will not rescue the proof strategy".[16][17][18]

## See also

- Arakelov theory

## References


1. Goldfeld, Dorian (1996). "Beyond the last theorem". *Math Horizons*. **4** (September): 26–34. doi:10.1080/10724117.1996.11974985 (https://doi.org/10.1080%2F10724117.1996.11974985). JSTOR 25678079 (https://www.jstor.org/stable/25678079).
2. Bombieri, Enrico (1994). "Roth's theorem and the abc-conjecture". *Preprint*. ETH Zürich.
3. Elkies, N. D. (1991). "ABC implies Mordell" (https://doi.org/10.1155%2FS1073792891000144). *International Mathematics Research Notices*. **1991** (7): 99–109. doi:10.1155/S1073792891000144 (https://doi.org/10.1155%2FS1073792891000144).
4. Pomerance, Carl (2008). "Computational Number Theory". *The Princeton Companion to Mathematics*. Princeton University Press. pp. 361–362.
5. Dąbrowski, Andrzej (1996). "On the diophantine equation $x! + A = y^2$". *Nieuw Archief voor Wiskunde, IV*. **14**: 321–324. Zbl 0876.11015 (https://zbmath.org/?format=complete&q=an:0876.11015).
6. Fesenko, Ivan (2015), "Arithmetic deformation theory via arithmetic fundamental groups and nonarchimedean theta functions, notes on the work of Shinichi Mochizuki" (https://www.maths.nottingham.ac.uk/personal/ibf/notesoniut.pdf) (PDF), *European Journal of Mathematics*, **1** (3): 405–440, doi:10.1007/s40879-015-0066-0 (https://doi.org/10.1007%2Fs40879-015-0066-0).

7. Oesterlé, Joseph (1988), "Nouvelles approches du "théorème" de Fermat" (http://www.numdam.org/item?id=SB_1987-1988__30__165_0), *Astérisque*, Séminaire Bourbaki exp 694 (161): 165–186, ISSN 0303-1179 (https://search.worldcat.org/issn/0303-1179), MR 0992208 (https://mathscinet.ams.org/mathscinet-getitem?mr=0992208)
8. Waldschmidt, Michel (2015). "Lecture on the abc Conjecture and Some of Its Consequences" (https://webusers.imj-prg.fr/~michel.waldschmidt/articles/pdf/abcLahoreProceedings.pdf) (PDF). *Mathematics in the 21st Century*. Springer Proceedings in Mathematics & Statistics. Vol. 98. pp. 211–230. doi:10.1007/978-3-0348-0859-0_13 (https://doi.org/10.1007%2F978-3-0348-0859-0_13). ISBN 978-3-0348-0858-3.
9. Elkies, N. D. (1991). "ABC implies Mordell" (https://doi.org/10.1155%2FS1073792891000144). *International Mathematics Research Notices*. **1991** (7): 99–109. doi:10.1155/S1073792891000144 (https://doi.org/10.1155%2FS1073792891000144).
10. Pomerance, Carl (2008). "Computational Number Theory". *The Princeton Companion to Mathematics*. Princeton University Press. pp. 361–362.
11. Pasten, Hector (2017), "Definability of Frobenius orbits and a result on rational distance sets", *Monatshefte für Mathematik*, **182** (1): 99–126, doi:10.1007/s00605-016-0973-2 (https://doi.org/10.1007%2Fs00605-016-0973-2), MR 3592123 (https://mathscinet.ams.org/mathscinet-getitem?mr=3592123), S2CID 7805117 (https://api.semanticscholar.org/CorpusID:7805117)
12. Ball, Peter (10 September 2012). "Proof claimed for deep connection between primes" (https://www.nature.com/news/proof-claimed-for-deep-connection-between-primes-1.11378). *Nature*. doi:10.1038/nature.2012.11378 (https://doi.org/10.1038%2Fnature.2012.11378). Retrieved 19 April 2020.
13. Revell, Timothy (September 7, 2017). "Baffling ABC maths proof now has impenetrable 300-page 'summary' " (https://www.newscientist.com/article/2146647-baffling-abc-maths-proof-now-has-impenetrable-300-page-summary/). *New Scientist*.
14. Conrad, Brian (December 15, 2015). "Notes on the Oxford IUT workshop by Brian Conrad" (https://mathbabe.org/2015/12/15/notes-on-the-oxford-iut-workshop-by-brian-conrad/). Retrieved March 18, 2018.
15. Castelvecchi, Davide (8 October 2015). "The biggest mystery in mathematics: Shinichi Mochizuki and the impenetrable proof" (https://doi.org/10.1038%2F526178a). *Nature*. **526** (7572): 178–181. Bibcode:2015Natur.526..178C (https://ui.adsabs.harvard.edu/abs/2015Natur.526..178C). doi:10.1038/526178a (https://doi.org/10.1038%2F526178a). PMID 26450038 (https://pubmed.ncbi.nlm.nih.gov/26450038).
16. Scholze, Peter; Stix, Jakob. "Why abc is still a conjecture" (https://web.archive.org/web/20200208075321/http://www.kurims.kyoto-u.ac.jp/~motizuki/SS2018-08.pdf) (PDF). Archived from the original (http://www.kurims.kyoto-u.ac.jp/~motizuki/SS2018-08.pd) on February 8, 2020. (updated version of their May report (http://www.kurims.kyoto-u.ac.jp/~motizuki/SS2018-05.pdf)|)
17. Klarreich, Erica (September 20, 2018). "Titans of Mathematics Clash Over Epic Proof of ABC Conjecture" (https://www.quantamagazine.org/titans-of-mathematics-clash-over-epic-proof-of-abc-conjecture-20180920/). *Quanta Magazine*.
18. "March 2018 Discussions on IUTeich" (http://www.kurims.kyoto-u.ac.jp/~motizuki/IUTch-discussions-2018-03.html). Retrieved October 2, 2018. Web-page by Mochizuki describing discussions and linking consequent publications and supplementary material

## Bibliography

- Lang, S. (1997), *Survey of Diophantine geometry* (https://books.google.com/books?id=b9RqCQAAQBAJ&pg=PA51), Berlin: Springer-Verlag, p. 51, ISBN 3-540-61223-8, Zbl 0869.11051 (https://zbmath.org/?format=complete&q=an:0869.11051)

- Szpiro, L. (1981). "Propriétés numériques du faisceau dualisant rélatif". *Seminaire sur les pinceaux des courbes de genre au moins deux* (http://www.numdam.org/item/AST_1981__86__R1_0.pdf) (PDF). Astérisque. Vol. 86. pp. 44–78. Zbl 0517.14006 (https://zbmath.org/?format=complete&q=an:0517.14006).
- Szpiro, L. (1987), "Présentation de la théorie d'Arakelov", *Contemp. Math.*, Contemporary Mathematics, **67**: 279–293, doi:10.1090/conm/067/902599 (https://doi.org/10.1090%2Fconm%2F067%2F902599), ISBN 9780821850749, Zbl 0634.14012 (https://zbmath.org/?format=complete&q=an:0634.14012)


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